



COFFEE
SKILLS
PROGRAM

Brewing

Intermediate

Practical Examination

Trainer Version (中文版)

Learner Name: _____

E-mail Address: _____

Learner No.: _____

Membership No.: _____

Learner's Country of Residence: _____

Trainer Name: _____ Exam Date: _____

Exam Information:

- No conferring is allowed, except where permitted in Section 3.
- Passing score on this exam is 70% overall.
- The learner has two attempts at the exam. If the learner does not pass after two attempts the learner, should take the course again.



Specialty
Coffee
Association



BREWING

部分	可获得总分	实际得分
第 1a 部分 - 萃取率 % 味道测试	12	
第 1b 部分 - 浓度 % 味道测试	12	
第 2 部分 - 冲煮测试	20	
第 3 部分 - 理想杯故障排除	20	
总分数	64	
最低合格分数: 45 分		

概述 Overview

实践考试的目的在于测试学生的以下能力: The intention of the practical exam is to test the student's ability to:

- 确定冲煮咖啡的浓度和萃取率差异; Identify strength and extraction differences in brewed coffee;
- 使用不同设备准备冲煮, 萃取率为 18% - 22%, 溶出物浓度至少为 1.15%, 以及; Prepare brews from different devices, each at 18% - 22% solubles yield, and each at a minimum of 1.15% solubles concentration, and;
- 修正自动过滤咖啡机冲煮失败的咖啡配方。Diagnose how to correct the recipe for a poorly brewed coffee from an Automatic Filter Brewer.

学生必须获得 70% 以上分数才能通过实践考试。The student must score at least 70% to pass the practical exam.

学生必须正确识别第 1a 部分中的至少一种冲煮, 还要正确识别第 1b 部分中的至少一种冲煮, 才能获得通过实践考试所需的最低总分。Students must correctly identify at least 1 brew in Part 1a, and also correctly identify at least 1 brew in Part 1b to achieve the minimum overall points necessary to pass this practical exam.

考试用咖啡 Coffee for Exams

第 1a 和 1b 部分所用咖啡必须是课堂活动和实践中使用过的咖啡, 学生应熟悉其参考冲煮。The coffee used in parts 1a and 1b must be a coffee that has been used for activities and practice during the class, and the student should be familiar with reference brews of this coffee.

第 2 和 3 部分所用咖啡必须是课堂活动或实践中未曾使用过的咖啡。The coffee used in parts 2 and 3 must be a coffee that has not been used for activities or practice during the class.

第 1 部分 - 味道测试 Part 1 - Taste tests

用于第 1a 和 1b 部分冲煮的咖啡在隔热玻璃瓶中存放不应超过 1 小时, 装杯之后, 不应冷却至室温。The coffees brewed for Parts 1a and 1b should be held in insulated flasks for no longer than 1 hour, and once served, should not be allowed to cool to room temperature.

第 1 部分 - 重考 Part 1 - Retakes

如果学生错误识别任何冲煮, 可以重考其一或两部分。可以在实践考试的任何其他测试前或后参加重考。The student may retake either or both parts if they incorrectly identify any brews. A retake may be taken before or after any other tests in this practical exam.

冲煮咖啡必须装入干净的杯子里, 用不同的代码标识以备重考, AST 必须随机排序冲煮咖啡, 因此顺序与第一次考试时不同。Brews must be served into clean cups with different codes for retakes, and the AST must randomise the order of the brews, so the order is different from the first attempt.

学生可于实践考试当天分别重考一次第 1a 部分及第 1b 部分。Students are permitted one retake for Part 1a, and one retake for Part 1b on the day of the practical exam.

如果学生重考第 1a 部分, 则将失去在第一次考试中获得的分。If a student attempts a retake for Part 1a, any points awarded in their first attempt will be forfeited.

如果学生重考第 1b 部分, 则将失去在第一次考试中获得的分。If a student attempts a retake for Part 1b, any points awarded in their first attempt will be forfeited.

第 1a 部分 - 萃取率 % 味道测试 Part 1a - Extraction % Taste Test

完成第 1a 部分的时间为 5 分钟。Students have 5 minutes to complete Part 1a.

AST 使用自动过滤咖啡机准备同一种咖啡的三种冲煮。The AST prepares three brews of the same coffee using an Automatic Filter Brewer.

所有冲煮咖啡均为 **1.30% 溶出物浓度**（容差 $\pm 0.05\%$ ），但**萃取率**各不相同：All brews must have a **1.30% solubles concentration** (with a tolerance of $\pm 0.05\%$), but each at different **solubles yield**:

- **16%、20% 和 24%**（容差 $\pm 1\%$ ）。

每种冲煮咖啡应存放于隔热玻璃瓶中，分别标识 16%、20% 和 24%。Each brew should be held in an insulated flask and labelled as 16%, 20%, and 24%.

将每种冲煮咖啡装入标有 3 位数代码的杯中。Each brew is served in a cup that is labelled with a 3-digit code.

- **第一次考试时为 744、921 和 680。** 744, 921, and 680 for the first attempt.
- **如需重考时，则为 049、478 和 815。** 049, 478, and 815 if a retake is required.

AST 必须留存单独记录，每个杯子装入的是哪种冲煮咖啡。The AST must keep a separate record of which brew is served in each cup.

第 1b 部分 - 浓度 % 味道测试 Part 1b - Strength % Taste Test

完成第 1b 部分的时间为 5 分钟。Students have 5 minutes to complete Part 1b.

AST 使用自动过滤咖啡机准备同一种咖啡的三种冲煮。The AST prepares three brews of the same coffee using an Automatic Filter Brewer.

所有冲煮咖啡均为 **20% 萃取率**（容差 $\pm 1\%$ ），但**溶出物浓度**各不相同：All brews must achieve a **20% solubles yield** (with a tolerance of $\pm 1\%$), but each with different **solubles concentrations**:

- **1.10%、1.30% 和 1.50%**（容差 $\pm 0.05\%$ ）。

每种冲煮咖啡应存放于隔热玻璃瓶中，分别标识 1.10%、1.30% 和 1.50%。Each brew should be held in an insulated flask and labelled as 1.10%, 1.30%, and 1.50%.

将每种冲煮咖啡装入标有 3 位数代码的杯中。Each brew is served in a cup that is labelled with a 3-digit code.

- **第一次考试时为 251、506 和 475。** 251, 506, and 475 for the first attempt.
- **如需重考时，则为 696、076 和 367。** 696, 076, and 367 if a retake is required.

AST 必须留存单独记录，每个杯子装入的是哪种冲煮咖啡。The AST must keep a separate record of which brew is served in each cup.

第 2 部分 - 冲煮测试 Brewing Test

学生可以选择培训中所使用的任何冲煮设备，自动冲煮设备**除外**。AST 和学生可以使用各种设备和方法来适应文化喜好，更具灵活性。Students have the option of using any brewing devices available at the campus, **except** any automatic brewing devices. This provides flexibility for ASTs and students to utilise a variety of devices and methods that suit cultural preferences.

在本章节提交的任何冲煮成品的最低溶出物浓度需为 1.15%。Any brews submitted in this section must have a minimum solubles concentration of 1.15%.

若学员自行选择过滤器材进行冲煮，则该过滤器材冲煮成品的萃取率需为 18%-22%。If the student chooses a drip filter device for any of the brews, the drip filter brews must achieve a solubles yield of 18% - 22%.

AST 须提供合适的冲煮比例和可供选择的冲煮设备，该冲煮比例由 AST 决定，冲煮成品需达到最低溶出物浓度 1.15%，并且在风味和口感上达到令人愉悦的平衡性。The AST must provide a suitable brew ratio to be used with alternative devices. This brew ratio is at the discretion of the AST and should be capable of producing a pleasant balance of taste and body when producing brews with a minimum solubles concentration of 1.15%.

若学员选择其他替代的过滤器材进行冲煮，仍必须参照使用 AST 规定的水粉比。If the student chooses an alternative device for any of the brews, the student must use the coffee to water ratio provided by the AST.

AST 应观察学生的冲煮过程，并就其所使用的技术予以反馈。The AST should observe the student during the brewing process and provide feedback on the techniques applied by the student.

AST 必须测试每次成功冲煮成品的样品，以确认该成品的溶出物浓度。The AST must test a sample of each successful brew to confirm the solubles concentration of that brew.

AST 必须计算过滤法冲泡的成功样本的萃取率。The AST must calculate the solubles yield of successful drip filter brews.

AST 必须审核完成冲煮后提交的工作表，并评估每份工作表是否填写正确。其中包括品尝相应的冲煮咖啡，并判断味道分析是否合理描述了冲煮咖啡。The AST must review the worksheets submitted for the completed brews and assess whether or not each worksheet has been completed correctly. This includes tasting the corresponding brew and determining if the taste analysis reasonably describes the brew.

请注意：要获得成功冲煮的 8 分，所提交的工作表无须正确。Note: Submitted worksheets do not have to be correct for 8 points to be awarded for the successful brew.

但是，工作表必须正确填写**所有**相关详细内容，才能获得正确填写工作表的 2 分。However, the worksheets must be correct, with **all** relevant details correctly completed, for 2 points to be awarded for correct worksheets.

第 2 部分 - 重考 Part 2 - Retakes

学员可以尝试重考第二部分。但是其将失去在第 2 部分第一次考试中获得的分數。Students may attempt a retake of Part 2. Any points awarded on the first attempt of Part 2 will be forfeited.

第 2 部分重考可以当场进行，或改天再考。The retake for Part 2 may be attempted immediately, or on another day.

第 3 部分 - 理想杯故障排除 Part 3 - Ideal Cup Troubleshooting

在考试的第 3 部分（理想杯故障排除）中，AST 必须准备一份不在“理想”范围内的自动过滤咖啡机冲煮的咖啡，使用 1:15 - 1:19 之间的粉水比，并要求学生在仅更改一个要素的前提下修正冲煮。In Part 3 of the exam (Ideal Cup Troubleshooting), the AST must prepare an Automatic Filter brew that is not in the “Ideal” range, using a coffee to water ratio between 1:15 - 1:19, and requires the student to make a change to only one element to correct the brew.

AST 必须**仅**通过以下要素之一，冲煮出“萃取不足”或“过度萃取”的咖啡：The AST must use **only one** of the following elements to produce either an under or over extracted brew:

- 研磨设置不当 Incorrect grind setting
- 水温不当 Incorrect water temperature
- 浸润时间（预浸润）不当 Incorrect wetting time (pre-wet)
- 给水时间不当 Incorrect water delivery time

AST 必须填写一份冲煮工作表，以便记录研磨设置、水温、浸润时间和给水时间，以供参考。The AST must complete a worksheet for the brew so that the grind setting, water temperature, wetting time, and water delivery time are recorded for reference.

建议 AST 在此故障排除测试中使用能产生明显“萃取不足”或“过度萃取”特征的咖啡，AST 意在使此咖啡的萃取率比其在均衡冲煮情况下的萃取率至少高或低 2%，并在“SCA 理想”范围之外。It is recommended that the AST uses a coffee in this troubleshooting test that is capable of producing clearly over or under extracted characteristics, and the AST aims for a solubles yield that is at least 2% more or less than a well-balanced brew of this coffee, as well as being outside the “SCA Ideal” range.

学生要在 30 分钟内完成以下任务：The student has 30 minutes to complete the following:

- 味道分析 Taste analysis
- 粉水比计算和折射仪分析 Coffee to water ratio calculation and refractometer analysis
- 诊断 Diagnosis
- 填写解决方案 Write a solution
- 准备修正后的冲煮。Prepare a corrected brew.

建议（但不作要求）学生使用工作表填写修正后的冲煮。It is recommended, but not required, that students use worksheets to complete the corrected brew.

AST 应在每位学生参加此部分考试之前校准折射仪，学生在使用折射仪时必须使用正确的样品测量方案。The AST should calibrate the refractometer prior to each student attempting this part of the exam, and students must use correct sample measuring protocols when using the refractometer.

学生必须使用与 AST 所用相似或相同的咖啡机准备修正后的冲煮。The student must prepare the corrected brew in the same or identical brewer used by the AST.

由 AST 对学生的冲煮咖啡进行测量，判断其是否符合“SCA 理想”范围。The student's brew is measured by the AST to determine if it is within the “SCA Ideal” range.

可在指定时间内准备多次冲煮。Multiple brews may be prepared within the allocated time.

如果自动过滤冲煮设备数量有限，学生可在任务 3.4 中结对准备修正后的冲煮。If automatic filter brewing equipment is limited, students may work in pairs to prepare the corrected brew in Part 3.4.

第 3 部分 - 计时 Part 3 - Timer

如果学生出于任何不公平的原因需要延迟完成任务，则 AST 有权自行决定停止和/或重置计时器。

(如：学生不得不等待设备可用。) The AST has the discretion to stop and/or reset the timer if the student is unfairly delayed in completing the Parts. (e.g. If the student has to wait for equipment to become available.)

如有必要(学生需要结对完成任务)，则计时器将在学生完成任务 3.3 之后到开始任务 3.4 之前的这段时间内停止。If necessary, the timer will be stopped between the completion of Part 3.3 and the beginning of Part 3.4 if students are asked to work in pairs.

第 3 部分 - 重考 Part 3 - Retakes

根据 AST 的决定，学生可以单独重考第 3 部分。如果学生要重考第 3 部分，则 AST 必须准备另一份不在“理想”范围内的自动过滤咖啡机冲煮的咖啡。AST 有权自行决定在重考过程中使用相同或不同的冲煮参数。At the discretion of the AST, a student may attempt a single retake of Part 3. If students retake Part 3, the AST must prepare another automatic filter brew that is not in the “Ideal” range. The AST has the discretion to use the same parameters or different parameters for the brew used in the retake.

第 3 部分重考可以当场进行，或改天再考。The retake for Part 3 may be attempted immediately, or on another day.

第 1 部分 - 味道测试 Part 1 - Taste tests

1a. 萃取率 % 味道测试: 1a. Solubles Yield % Taste Test:

这三个带编号的杯子里分别装有同一种咖啡的不同冲煮。Each of these three numbered cups contains a different brew of the same coffee.

溶出物浓度均为 1.30%，但萃取率分别为：All three have a 1.30% solubles concentration, but each has a different solubles yield:

- 16%、20% 和 24%。

请品尝每一杯，并确定每一杯的萃取率 %。Please taste each one and identify which solubles yield % matches each cup.

- 完成此部分的时间为 5 分钟。You have 5 minutes to complete this section.
- 必须正确识别至少一个样品。You must correctly identify at least one sample.

如果您正确识别一杯的萃取率，则可获得 4 分。If you correctly identify one cup, you will be awarded 4 points.

如果您 3 杯全部识别正确，则可获得 12 分。If you correctly identify all 3 cups, you will be awarded 12 points.

如有必要，您可在参加本次实践考试的其他部分前或后重考此部分。If necessary, you may retake this section before or after attempting other sections in this practical exam.

在您重考之前，样品将被重新标识并重新随机排序。The samples will be relabelled and re-randomised before you attempt a retake.

如果您重考第 1a 部分，则将失去在第一次考试中获得的分數。If you attempt a retake of Part 1a, you will lose any points awarded in your first attempt.

统一的溶出物浓度: 1.30%			
冲煮	确定冲煮 (圈出冲煮的编号)	第一次考试	重考
		正确 是/否	正确 是/否
16% 的萃取率	744 - 921 - 680		
	049 - 478 - 815		
20% 的萃取率	744 - 921 - 680		
	049 - 478 - 815		
24% 的萃取率	744 - 921 - 680		
	049 - 478 - 815		
正确识别的冲煮个数			
正确识别 1 个可获得 4 分 正确识别 3 个可获得 12 分 (最多可获得 12 分)		实际得分	

1b.溶出物浓度 % 味道测试: Solubles Concentration % Taste Test:

这三个带编号的杯子里分别装有同一种咖啡的不同冲煮。Each of these three numbered cups contains a different brew of the same coffee.

萃取率均为 20%，但溶出物浓度分别为: All three have a 20% solubles yield but each has a different solubles concentration

- 1.10%、1.30% 和 1.50%。

请品尝每一杯，并确定每一杯的溶出物浓度 %。Please taste each one and identify which solubles concentration % matches each cup.

- 完成此部分的时间为 5 分钟。You have 5 minutes to complete this section.
- 必须正确识别至少一个样品。You must correctly identify at least one sample.

如果您正确识别一杯的萃取率，则可获得 4 分。If you correctly identify one cup, you will be awarded 4 points.

如果您 3 杯全部识别正确，则可获得 12 分。If you correctly identify all 3 cups, you will be awarded 12 points.

如有必要，您可在参加本次实践考试的其他部分前或后重考此部分。If necessary, you may retake this section before or after attempting other sections in this practical exam.

在您重考之前，样品将被重新标识并重新随机排序。The samples will be relabelled and re-randomised before you attempt a retake.

如果您重考第 1b 部分，则将失去在第一次考试中获得的分數。If you attempt a retake of Part 1b, you will lose any points awarded in your first attempt.

统一的萃取率： 20%			
冲煮	确定冲煮 (圈出冲煮的编号)	第一次考试	重考
		正确 是/否	正确 是/否
1.10% 的溶出物浓度	251 - 506 - 475		
	696 - 076 - 367		
1.30% 的溶出物浓度	251 - 506 - 475		
	696 - 076 - 367		
1.50% 的溶出物浓度	251 - 506 - 475		
	696 - 076 - 367		
正确识别的冲煮个数			
正确识别 1 个可获得 4 分 正确识别 3 个可获得 12 分 (最多可获得 12 分)		实际得分	

2. 冲煮测试 Brewing Test

完成以下任务的时间为 40 分钟。You have 40 minutes to complete the following Parts.

请选择培训中所使用的任意 2 种冲煮设备（自动咖啡机除外）。Please choose any 2 brewing devices that are available at this campus (except an automatic brewer).

请分别使用每种设备准备一份冲煮，溶出物浓度最低为 1.15%。Please prepare one brew from each device that achieves a minimum solubles concentration of 1.15%.

如果您选择了过滤式冲煮，则该冲煮成品的萃取率需为18%-22%。If you choose a drip filter brewer, the brew must also achieve a solubles yield of 18% to 22% (a successful brew).

如果您选择了替代的冲煮器材，则必须参照使用AST规定的水粉比，且冲煮成品的最低溶出物浓度需为 1.15%。If you choose an alternative device, you must use a coffee to water ratio within the range described by the AST, and the brew must achieve a solubles concentration of 1.15% (a successful brew).

您可以分别使用每种设备准备多次冲煮，但仅可提交其中一份冲煮进行评分。Multiple brews may be prepared with each device, but only one brew from each device may be submitted for scoring.

请为在此部分准备的每一份冲煮分别填写一份工作表（包括折射仪分析）提交相应的工作表后，方可获得成功冲煮的分数。You are required to complete a worksheet (including refractometer analysis) for each brew you prepare in this section. A successful brew will not receive points unless you submit a corresponding worksheet.

如果您使用第一台设备成功冲煮，并提交了相应的工作表，则可获得 8 分。如果您使用此设备成功冲煮，并正确填写工作表，则可额外获得 2 分。If you submit a successful brew from the first device, and a corresponding worksheet, you will be awarded 8 points. If your brew from this device is successful, and your worksheet is completed correctly, you will be awarded an additional 2 points.

如果您使用第二台设备成功冲煮，并提交了相应的工作表，则可获得 8 分。如果您使用此设备成功冲煮，并正确填写工作表，则可额外获得 2 分。If you submit a successful brew from the second device, and a corresponding worksheet, you will be awarded 8 points. If your brew from this device is successful, and your worksheet is completed correctly, you will be awarded an additional 2 points.

只有当成功冲煮时，才能获得正确填写工作表的分数。Points can only be awarded for correct worksheets if their corresponding brew is successful.

评估人将品尝每份冲煮，比较其味道分析；并会测量每份冲煮样品，确认其折射仪分析是否正确。The AST will taste each brew to compare taste analysis, and the assessor will measure a sample of each brew to confirm correct refractometer analysis.

您必须在 40 分钟的时限内成功冲煮并填写相应的工作表。Successful brews and their corresponding completed worksheets must be submitted within the 40-minute time limit.

如发现设备异常，请告知评估人，其可酌情停止计时。If you believe any of the equipment is not functioning correctly, please inform the assessor who may stop the time if appropriate.

如有必要，评估人还可重启测试，并酌情重置计时器。If necessary, the Part may be restarted, and the timer reset appropriately.

冲煮测试				可获得分数	实际得分
冲煮 1	溶出物浓度	萃取率	冲煮达成目标，并提交了工作表 是/否	8	
	%	%	工作表填写正确 是/否	2	
冲煮 2	溶出物浓度	萃取率	冲煮达成目标，并提交了工作表 是/否	8	
	%	%	工作表填写正确 是/否	2	
完成冲煮和工作表的时间：_____:					
总得分					

3. 理想杯故障排除测试 Ideal Cup Troubleshooting Test

完成以下任务的时间为 30 分钟。You have 30 minutes to complete the following tasks.

在任务 3.1、3.2, 和 3.3 中, 您**不可**与其他学生商量。In Part 3.1, 3.2, and 3.3 you may **not** consult with other students.

如有必要, 则计时器将在您完成任务 3.3 之后到开始任务 3.4 之前的这段时间内停止。If necessary, the timer will be stopped between the completion of Part 3.3 and the beginning of Part 3.4.

如发现设备异常, 请告知评估人, 其可酌情停止计时。If you believe any of the equipment is not functioning correctly, please inform the assessor who may stop the time if appropriate.

如有必要, 评估人还可重启测试, 并酌情重置计时器。If necessary, a task may be restarted, and the timer reset appropriately.

任务 3.1

评估人使用自动过滤咖啡机冲煮了咖啡, 因为是有意为之, 所以咖啡萃取率低于 18% (萃取不足) 或 高于 22% (过度萃取)。The AST has brewed a coffee with an automatic filter brewer, and intentionally brewed the coffee so that it has achieved either a solubles yield below 18% (under extracted), or a solubles yield above 22% (over extracted).

您必须品尝并说明冲煮咖啡属于萃取不足还是过度萃取。You must taste the coffee and indicate whether you think the brew is under extracted, or over extracted.

如果您的味道分析正确, 则可获得 4 分。If your taste analysis is correct, you will be awarded 4 points.

任务 3.2

评估人会告诉您用于准备此次冲煮的咖啡粉量和用水量。您必须据此计算此次冲煮的粉水比, 并使用折射仪测量冲煮样品, 判断溶出物浓度。The AST will then tell you the ground coffee and water doses used to prepare the brew. You must then calculate the coffee to water ratio used for the brew, and then measure a sample of the brew with a refractometer to determine the solubles concentration.

必须遵循正确的样品测量方案。Correct sample measuring protocols must be followed.

使用该粉水比和溶出物浓度, 并结合冲煮品管表, 判断萃取率。You must then use the coffee to water ratio and the solubles concentration to determine the solubles yield with the brewing control chart.

如果您的计算正确, 并正确使用折射仪和冲煮品管表, 则可获得 4 分。If your calculations and use of the refractometer and brewing control chart are correct, you will be awarded 4 points.

任务 3.3

您必须检查自动过滤咖啡机和研磨机, 并诊断造成萃取不足或过度萃取的原因。如果您的诊断正确, 则可获得 4 分。You must then examine the automatic filter brewer and grinder and diagnose the cause of the brew being under or over extracted. If your diagnosis is correct, you will be awarded 4 points.

必须描述修正冲煮的解决方案。Then you must describe a solution to correct the brew.

如果您的解决方案正确, 则可获得 4 分。If your solution is correct you will be awarded 4 points.

如果您的诊断和解决方案均正确, 则可进入任务 3.4。If your diagnosis and solution are correct, you may proceed to Part 3.4.

否则不得进入任务 3.4。If your diagnosis and solution are not correct, you may not proceed to Part 3.4.

任务 3.4

现在您必须使用您的解决方案, 在同一台自动过滤咖啡机上准备冲煮, 并且溶出物浓度为 1.15% - 1.45%, 萃取率为 18% - 22%。必须使用与评估人所用相同的粉水比。You must now use your solution to prepare a brew in the same automatic filter brewer, and achieve a solubles concentration of 1.15% to 1.45%, and a solubles yield of 18% to 22%. You must use the same coffee to water ratio as was used by the AST.

您可以准备多次冲煮来获得目标冲煮, 但是目标冲煮必须在指定时间内完成。You may prepare multiple brews to achieve the target brew, but the target brew must be completed within the allocated time.

如果您在指定时间内成功冲煮, 则可获得 4 分。If you prepare a successful brew within the allocated time, you will be awarded 4 points.

理想杯故障排除测试				可获得 分数	实际得分
任务 3.1 味道分析: 描述冲煮。 (选择其一)	萃取不足 或 过度萃取		描述正确 是/否	4	
任务 3.2 折射仪分析: 计算粉水比, 并正 确测量和绘制冲煮 表。	粉水比 1: _____		正确计算粉水比, 正确使用折射仪, 正确使用冲煮品管 表。 是/否	4	
	溶出物浓度 _____ %	萃取 率 _____ %			
任务 3.3 诊断和 修正冲煮的解决方 案。	冲煮不当的原因?			4	
	描述		描述正确 是/否		
	修正冲煮的解决方案?			4	
	解决方案		解决方案得当 是/否		
学生在任务 3 中的诊断正确并且解决方案得当, 方可进入任务 3.4					
任务 3.4 学生冲煮的 浓度 和萃取率。	溶出物浓度 _____ %	萃取 率 _____ %	“理想” 范围 是/否	4	
				总得分	